

Scientific Consensus Report

RESEARCH QUERY:

Does metformin improve survival in HER2-positive breast cancer?

1. Executive Consensus & Synthesis

The Answer is: A comprehensive scientific consensus report cannot be generated without specific paper details and relationships provided in the prompt. Please provide the necessary evidence and connections for a detailed assessment of clinical utility and prognosis in HER2-positive breast cancer patients treated with metformin.

2. Laboratory Protocol Draft

****Laboratory Experiment Protocol: Evaluating the Efficacy of Metformin in HER2-positive Breast Cancer Cells****

****1. EXPERIMENTAL HYPOTHESIS:****

The primary hypothesis being tested is that metformin treatment enhances survival and inhibits proliferation of HER2-positive breast cancer cells.

****2. MATERIALS & REAGENTS:****

* Primary cell lines:

- + MDA-MB-231 (HER2-positive breast cancer cell line)
- + MCF-7 (HER2-negative breast cancer cell line, as a control)

* Drugs and compounds:

- + Metformin hydrochloride (Sigma-Aldrich, 100mM stock solution)
- + Azacitidine (Sigma-Aldrich, 10mM stock solution)
- + DMSO (vehicle control)

* Assay kits and tools:

- + Cell viability assay (e.g., MTT or Alamar Blue)
- + Flow cytometry reagents for apoptosis analysis
- + PIK3CA mutation-specific primers for genotyping (optional)

* Other materials:

- + Cell culture media and supplements
- + Fetal bovine serum (FBS)
- + Phosphate-buffered saline (PBS)

****3. STEP-BY-STEP PROCEDURE:****

Step 1: Cell Culture and Treatment

1. Seed MDA-MB-231 cells at a density of 1×10^5 cells/mL in 96-well plates.
2. Allow the cells to attach for 24 hours before treating with metformin or vehicle control (DMSO).
3. Treat cells with 0, 5, 10, and 20 mM metformin, respectively, for 48 hours.

Step 4: Cell Viability Assay

1. Perform a cell viability assay using MTT reagent.
2. Measure the absorbance at 570 nm using a microplate reader.

Step 5: Flow Cytometry Analysis of Apoptosis

1. Harvest cells treated with metformin or vehicle control.
2. Stain cells with Annexin V-FITC and PI.
3. Analyze samples by flow cytometry to detect early apoptosis (Annexin V+).

Step 6: Genotyping for PIK3CA Mutations (Optional)

1. Harvest cells treated with metformin or vehicle control.
2. Perform PCR amplification of the PIK3CA gene using mutation-specific primers.
3. Sequence amplified products to determine PIK3CA mutation status.

4. CONTROLS

- * Negative Control: Untreated MDA-MB-231 cells
- * Positive Control: Cells treated with a known chemotherapy agent (e.g., doxorubicin) or apoptosis inducer (e.g., staurosporine)

5. SAFETY & PRECAUTIONS

- * Handle metformin and other reagents in a well-ventilated area, using gloves and protective eyewear.
- * Avoid exposure to DMSO by wearing gloves when handling it.
- * Follow standard protocols for cell culture and flow cytometry procedures.

Additional Considerations:

- * Ensure that the cells are properly authenticated and tested for mycoplasma contamination before conducting experiments.
- * Use a water bath or thermostatically controlled incubator to maintain consistent temperatures during treatments.
- * Conduct experiments in triplicate to ensure reliable results.

3. Cited Evidence & Methodological Quality

[1] Optimizing clinical monitoring and management guidelines for capivasertib in HR-positive/HER2-negative advanced breast cancer: expert opinion.

Score:

[2] Therapy as a State-Generator: Dynamic Phenotypic Landscapes and Adaptive Stress Circuits in Chemotherapy Resistance of Breast Cancer.

Score:

[3] Window-of-opportunity trials to screen effective agents and optimize dose in breast cancer prevention.

Score:

[4] Combined anti-tumor effects of metformin and azacitidine in breast cancer cells.

Score:

[5] Non-coding RNA-mediated mechanisms underlying tamoxifen resistance in breast cancer.

